ECET 250E-Linear Circuit 1
Problemset 5-Chapter 6- Capacitors and Inductors

Name: _____ Marks: _____

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1. **6.16** The equivalent capacitance at terminals *a*-*b* in the circuit of **Fig. 6.50** is 30 μF. Calculate the value of *C*.

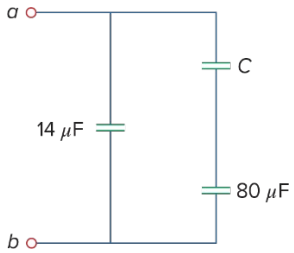


Figure 6.50 For **Prob. 6.16**.

2. **6.17** Determine the equivalent capacitance for each of the circuits of **Fig. 6.51**.

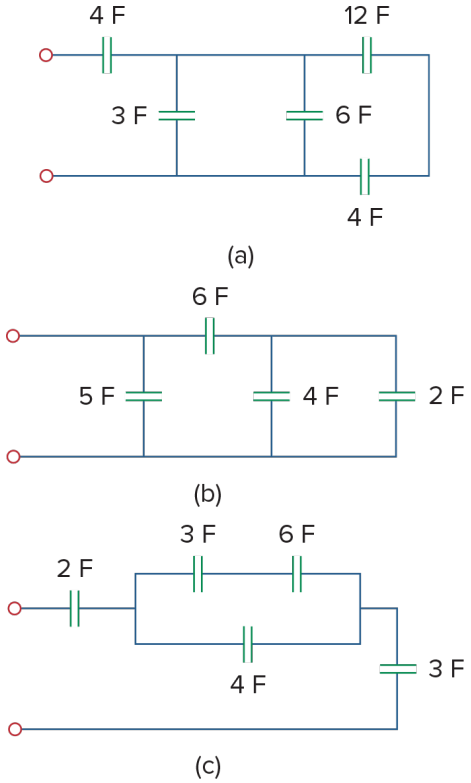


Figure 6.51 For **Prob. 6.17**.

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

3. **6.18** Find Ceq in the circuit of **Fig. 6.52** if all capacitors are 4 μF.

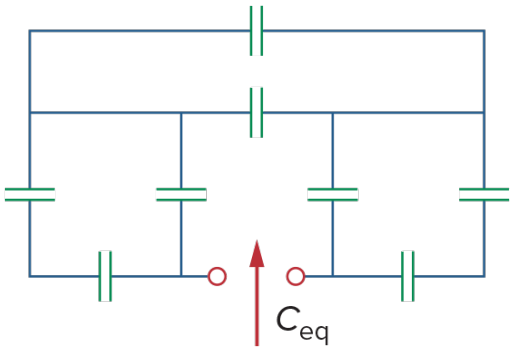


Figure 6.52 For **Prob. 6.18**.

4. **6.26** Three capacitors, C1=5 μF, C2=10 μF, and C3=20 μF, are connected in parallel across a 150-V source. Determine:
   1. (a)the total capacitance,
   2. (b)the charge on each capacitor,
   3. (c)the total energy stored in the parallel combination

5. **6.32** In the circuit in **Fig. 6.64**, let is=$50e^{-2t}$ mA and $v_1(0)$=50 V, $v_2(0)$=20 V. Determine: (a) $v_1(t)$ and $v_2(t)$, (b) the energy in each capacitor at t=0.5 s.

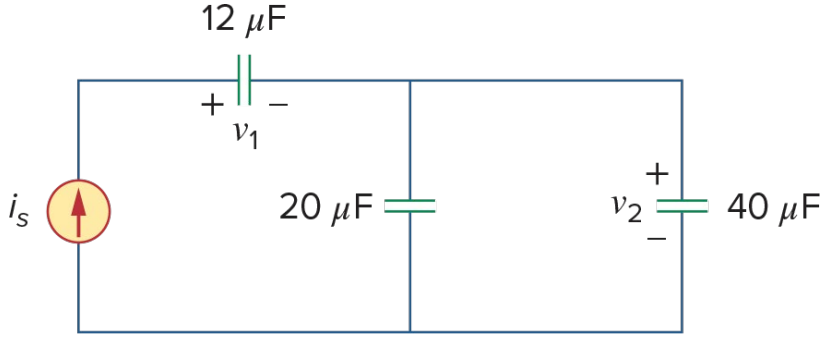


Figure 6.64 For **Prob. 6.32**.

6. **6.40** The current through a 5-mH inductor is shown in **Fig. 6.66**. Determine the voltage across the inductor at t=1, 3, and 5 ms.

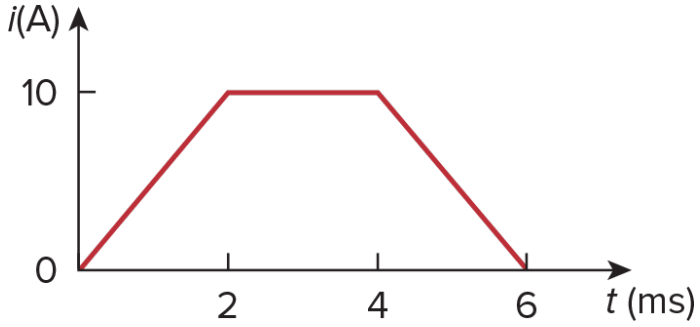


Figure 6.66 For **Prob. 6.40**.

7. **6.43** The current in an 80-mH inductor increases from 0 to 60 mA (steady state). How much energy is stored in the inductor?

8. **6. 51** Determine Leq at terminals *a*-*b* of the circuit in **Fig. 6.73**.

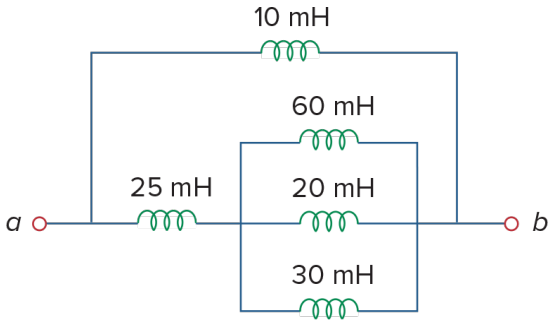


Figure 6.73 For **Prob. 6.51**.

9. **6.53** Find Leq at the terminals of the circuit in **Fig. 6.75**.

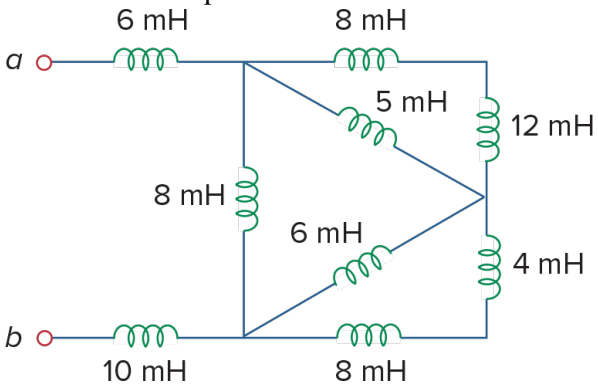


Figure 6.75 For **Prob. 6.53**.

10. 6.60 In the circuit of **Fig. 6.82**, io(0)=2 A. Determine io(t) and vo(t) for t>0.

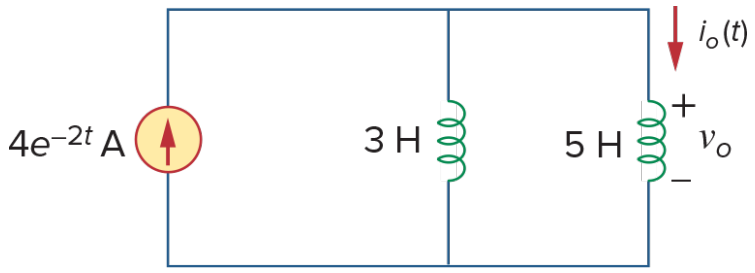


Figure 6.82 For **Prob. 6.60**.